

JPE Replication Report

PUBLISHED

September 13, 2026

1 Paper Identification

ID	20221230
Author	Mitman
Title	Expectation and Wealth Heterogeneity in the Macroeconomy
Iteration	1

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

1. You refer to the replication package of Krueger, Dirk, Kurt Mitman, and Fabrizio Perri. 2016. "Macroeconomics and Household Heterogeneity." In *Handbook of Macroeconomics*. I could not find that package anywhere [here](#). If you want to use that as the source of the raw data, i need a DOI for it (package needs to be published on a [trusted repository](#)). Regardless of whether you want to refer to this package or not, I doubt this is within the PSID license to start with. The point of that license is that everybody may use PSID data, but everyone needs to be registered *before* that, i.e. each new user needs to agree to terms and conditions. You are *not allowed* to post any PSID data in any replication package. I am afraid I need to ask you to create a dedicated PSID deposit for this purpose. You can see our guidance for this [here](#). I can see that this is annoying, but we are getting frequent takedown requests from PSID because of that very reason, and the only thing I can assure is that the pain is larger, the more time has passed.
2. We are struggling the cover the entire computational burden. In fact, we were only to get a very small part of the pipeline. We seem to have run out of RAM at 256GB and many hours runtime. So, I'm asking you for a way to abbreviate that pipeline, with you sharing with us some precomputed results. I would be happy with reproducing the baseline and 2 or 3 of your scenarios.
3. There are a few other minor comments below about readme issues, please have a look at those.

3 General

- ☒ Data Availability and Provenance Statements
 - ☒ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☒ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☒ License for Data (optional, but recommended)
 - ☒ Details on each Data Source
- ☒ Dataset list
- ☒ Computational requirements

- ☒ Software Requirements
- ☒ Controlled Randomness (as necessary)
- ☒ Memory, Runtime, and Storage Requirements
- ☒ Description of programs/code
 - ☒ License for Code (Optional, but recommended)
 - ☒ Details (as necessary)
- ☒ List of tables and programs
- ☒ References (Optional, but recommended)

4 High-level Description of Research Project

This research project...

- ☒ uses observational data.
- ☐ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
 - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☐ is a simulation study: the data are generated by an algorithm.
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

5.1 Data Sources

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
<code>_data/_data/FRBNY-SCE-Public-Microdata-Complete-13-16.csv</code> ; <code>...17-19.csv</code> ; <code>...latest.csv</code>	yes	no	yes	yes	yes	no
<code>_data/_data/SCE_public_HH-finance_quarterly_microdata.xlsx</code>	yes	no	yes	yes	yes	no
<code>_data/_data/PROB_SPF_RAW.xlsx</code> ; <code>_data/_data/PROB_SPF_RAW_INF.xlsx</code>	yes	no	yes	yes	yes	no
<code>_data/_data/CPIAUCSL.xls</code> ; <code>CPILFESL.xls</code> ; <code>CSUSHPINSA.xls</code> ; <code>UNRATE_prob.xls</code>	yes	no	yes	yes	yes	no

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
_data/_data/_data_var/*.xls; _data/_data/business_cycle_data.xlsx	yes	no	yes	yes	yes	no
_data/_data/psid_2004_2010.dta; _data/_data/psid_2004_2010.csv	yes	no	yes	yes	yes	no
CPS calibration targets in _model/_aux/_baseline/modelsMasterExcel.xlsx	no	no	yes	yes	yes	no

Paper included in the package does not contain data citations.

6 Requirements

- ☒ README is in TXT, MD, or PDF format
- ☒ README is accessible at the root of the package (not inside a folder)
- ☐ Title conforms to guidance (starts with “Data and Code for: Title of Paper” or “Code for: Title of Paper”, is properly capitalized)
- ☒ Authors (with affiliations) are listed in the same order as on the paper

{REQUIRED} Please review the title of the deposit as per our guidelines.

6.1 Stated Requirements

- ☐ No requirements specified
- ☒ Software Requirements specified as follows:
 - macOS
 - R 4.5.1
 - Stata/MP 17.0
 - MATLAB R2024b with Econometrics, Statistics and Machine Learning, Optimization, Symbolic Math, and Parallel Computing Toolboxes
- ☒ Computational Requirements specified as follows:
 - 64 GB RAM
 - 100 GB free storage
- ☒ Time Requirements specified as follows:
 - Data branch: 20–40 minutes
 - Model branch: 60–70 hours;
- ☐ Requirements are complete.

Authors are invited to specify the macOS version.

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals.* If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary: - Data files with PII indicators: 6 - Variables flagged in data: 35 - Code files with PII references: 94 - PII references in code: 1436

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	FRBNY-SCE-Public-Microdata-Complete-13-16.csv	4	lat
Data	FRBNY-SCE-Public-Microdata-Complete-17-19.csv	4	lat
Data	FRBNY-SCE-Public-Microdata-latest.csv	4	lat
Data	LICENSE.txt	1	son
Data	psid_2004_2010.csv	6	house, phone, sex, child, wife
Data	psid_2004_2010.dta	16	house, phone, sex, child, wife, school
Code	01_paper.tex	628	lat, lon, name, loc, location, house, sex, gender, second, son, address, degree, city, compound, spouse
Code	BKKS_GIRFs.m	125	son, lat, loc, location, name
Code	BKKS_load_parameters.m	6	loc, location, name
Code	BKKS_load_parameters.m	6	loc, location, name
Code	BKKS_main.m	8	lat, lon
Code	BKKS_main.m	11	lat, lon, second
Code	BKKS_start_runs.m	7	loc, location
Code	BKKS_start_runs.m	54	loc, location
Code	DoMomentsTouchup.m	1	lat
Code	DoPanelExPost.m	2	lat

File Type	File	Variables/References	PII Categories
Code	GaussHermite_2.m	1	degree
Code	InfoDecomposition.m	4	name, lat
Code	PanelForAlex.m	4	lat
Code	Redo_Wealth_Moments.m	1	loc, location
Code	Redo_Wealth_Moments_Ent.m	1	loc, location
Code	bvar_data.R	8	name
Code	bvar_est.m	3	name, lat
Code	calculate_fcerrors_stoch.m	2	lat
Code	calculate_fcerrors_stoch.m	2	lat
Code	calculate_fcerrors_stoch_optimized.m	5	lat
Code	calculate_ineq_moments_stoch.m	3	lat
Code	calculate_ineq_moments_stoch.m	3	lat
Code	calculate_moments_stoch.m	3	lat
Code	calculate_moments_stoch.m	6	lat
Code	data_bib.tex	30	name, url, house, social, census, lat, city, son
Code	data_master.R	5	loc
Code	figure_1_ab_table_a7a9.R	2	lat, name
Code	figure_1_cd.R	2	lat, name
Code	figure_2_a.R	2	name
Code	figure_2_b.R	2	name
Code	figure_3_and_4.m	36	name, lat
Code	figure_5.m	19	name, lat
Code	figure_6.m	3	name, lat
Code	figure_6_data.R	2	name
Code	figure_7.m	31	name, loc, location, son, lat
Code	figure_8.m	9	name, lat

File Type	File	Variables/References	PII Categories
Code	<code>figure_b1.m</code>	3	name, loc, location, lat
Code	<code>figure_b2.m</code>	3	name, lat
Code	<code>figure_c1.m</code>	27	name, lat
Code	<code>figure_c2.m</code>	2	name, loc, location
Code	<code>figure_c3.m</code>	2	name, loc, location
Code	<code>figure_c4.m</code>	3	name
Code	<code>figure_d1.m</code>	17	name, lat
Code	<code>getDirs.m</code>	1	name
Code	<code>get_moments_table.m</code>	1	lname, name
Code	<code>get_moments_table.m</code>	1	lname, name
Code	<code>get_moments_table.m</code>	1	lname, name
Code	<code>get_moments_table2.m</code>	1	lname, name
Code	<code>inequality_moments.m</code>	41	lat, name
Code	<code>make_readme_pdf-checkpoint.py</code>	17	lat, url, lon, zip, loc, name
Code	<code>make_shocks.m</code>	3	lat, minute, name
Code	<code>make_table3_data.do</code>	34	house, lat, loc, name
Code	<code>make_table_bench_compare.m</code>	20	name, lat
Code	<code>make_table_entrepreneur_compare.m</code>	23	name, lat
Code	<code>make_table_entrepreneur_fi.m</code>	1	lat
Code	<code>model_master.m</code>	4	lat, name
Code	<code>packages_rstudio.R</code>	2	lat, name
Code	<code>rouwenhorst.m</code>	3	lon, lat
Code	<code>run_all.sh</code>	22	loc, name, minute
Code	<code>sce_data.R</code>	22	name, lat, sex
Code	<code>simKS.m</code>	1	son
Code	<code>simKS.m</code>	1	son
Code	<code>slim_models_tmp.m</code>	31	name, block, loc, lat

File Type	File	Variables/References	PII Categories
Code	<code>solve_EGM.m</code>	4	lat, lon
Code	<code>solve_EGM.m</code>	4	lat, lon
Code	<code>solve_experience.m</code>	1	lat
Code	<code>solve_experience.m</code>	11	lat, loc, block
Code	<code>spf_data.m</code>	4	name, second, lat
Code	<code>table_a10.R</code>	3	lat, name
Code	<code>table_a10.tex</code>	1	lat
Code	<code>table_a8.R</code>	6	name, lname, city, lat, house
Code	<code>table_a8.tex</code>	2	lat, house
Code	<code>table_b2_a.m</code>	5	name
Code	<code>table_bii.m</code>	10	name, loc, location
Code	<code>table_ci.m</code>	7	name, loc, location
Code	<code>table_cii.m</code>	2	name, lat
Code	<code>table_ciii.m</code>	4	name
Code	<code>table_civ.m</code>	2	name, lat
Code	<code>table_cix.m</code>	2	name, lat
Code	<code>table_cv.m</code>	2	name, lat
Code	<code>table_cvi.m</code>	2	name, lat
Code	<code>table_cvii.m</code>	2	name, lat
Code	<code>table_cviii.m</code>	2	name, lat
Code	<code>table_cx.m</code>	2	name, lat
Code	<code>table_cxi.m</code>	4	name
Code	<code>table_cxii.m</code>	2	name, lat
Code	<code>table_di.m</code>	2	name, lat
Code	<code>table_i.tex</code>	1	lat
Code	<code>table_i_and_ii.m</code>	4	name, lat
Code	<code>table_iii.m</code>	3	name
Code	<code>table_iv.m</code>	2	name, lat

File Type	File	Variables/References	PII Categories
Code	table_vi.m	2	name, lat
Code	wprctile.m	7	lat, lon
Code	wprctile.m	7	lat, lon

See [Appendix](#) for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-08-15T19:44:09.240

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any windows filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that STATA allows / to be a filepath separator on a windows platform, hence this is a requirement for all STATA applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (C:\ etc)
254	12	40	44	2

7.1.3 Duplicate Files Report

We found the following duplicate files:

Filename	Size (MB)	Checksum (MD5)
/__MACOSX/_replication/data/.DS_Store	0.0	0bed7e90c2bade9763fa18f1fb4441d31f91c87c
/__MACOSX/_replication/model/.DS_Store	0.0	0bed7e90c2bade9763fa18f1fb4441d31f91c87c
/__MACOSX/_replication/_model/aux/.DS_Store	0.0	0bed7e90c2bade9763fa18f1fb4441d31f91c87c
/__MACOSX/_replication/paper/.DS_Store	0.0	0bed7e90c2bade9763fa18f1fb4441d31f91c87c
/_replication/_model/_aux/_kvar/basefun.m	0.0	06c2e6b175910e0bd168d593f7a36d5cd2bed34e
/_replication/_model/_aux/_kvar/BKKS_load_parameters.m	0.0	70bf4de76c6edce4980a96c671a4f09a03ccc5f0
/_replication/_model/_aux/_kvar/genShocks.m	0.01	c4552357e0a7c2362079a63962c3dc3540321e72
/_replication/_model/_aux/_kvar/wprctile.m	0.01	571c12e79199a7348b916b5cac28bc7cee415aac
/_replication/_model/_aux/_kvar/calculate_fcerrors_stoch.m	0.02	60f4c7288cade86c073a2c302578c55462d32c23

7.1.4 Zero Size Files Report

We found the following zero size files:

Filename	Size (MB)	Checksum (MD5)
/_replication/.Rhistory	0.0	da39a3ee5e6b4b0d3255bfef95601890afd80709

7.1.5 Large Files Report

We did not find any files larger than 100MB.

7.2 Code description

The replication package contains two main branches: an R/Stata _data pipeline and a MATLAB _model pipeline. The data branch reproduces Section 2, Online Appendix A, Table B.2 panel (a), and Table III’s data column. The model branch solves the quantitative model and generates the remaining paper and appendix tables/figures.

cloc		github.com/AIDanial/cloc v 2.02 T=0.46 s (293.0 files/s, 394373.3 lines/s)		
Language	files	blank	comment	code
CSV	6	0	0	158748
MATLAB	82	2971	2827	8875
TeX	27	585	62	4765
R	11	309	60	1090
Markdown	1	131	0	536
Python	1	32	55	192
Stata	1	75	3	158
Bourne Shell	1	28	42	134
Text	5	8	0	33
---	---	---	---	---
SUM:	135	4139	3049	174531

☒ The replication package contains a "main" or "master" file(s) which calls all other auxiliary programs.

7.3 Computing Environment of the Replicator

- Nuvolos Cloud, Ubuntu 22.04.2 LTS, 16-core CPU, 64 GB RAM (up to 256GB for model part), 100 GB storage
- Stata/MP 18.0
- Matlab R2024a
- R 4.5.1.

8 Replication

- Downloaded code and data from Dropbox.
- As the replication setup does not allow for easy cross-application interactions, replication was performed manually by branch, without the use of `run_all.sh`.
- Running the main file `data_master.R`, it seems that the R scripts sourced were meant to be run in separate environment, as inner scripts replace global variables or clean the whole environment. However, the current version of the code, which uses `source()`, runs those script in the global environment leading to issues similar to the one reported below.
- Example issue: the WD path seems to be overwritten by `_aux/sce_data.R`.

```
source("figure_1_ab_table_a7a9.R") # Figure 1a, 1b; Tables A.7, A.9
Error in file(filename, "r", encoding = encoding) :
  cannot open the connection
In addition: Warning message:
In file(filename, "r", encoding = encoding) :
  cannot open file 'figure_1_ab_table_a7a9.R': No such file or directory

> WD
[1] "/files/_replication/_data/_aux"
```

These issues were solved by renaming the variable `WD` in `data_master.R` to `MASTER_WD` to avoid it being overwritten in subscripts and by adding the `local = TRUE` option to `source`.

- After completion of the data part, launched `_model/model_master.m`. Running this required more than RAM (> 128GB) on our setup than stated by authors.
- Replicated baseline scenarios 1001, 1007, 1018, 1023, 1025. Manually interrupted replication due to considerable resource consumption.

9 Findings

9.1 Missing Requirements

- ☐ Software Requirements
 - ☐ OS version

[REQUIRED] Please amend README to contain complete requirements.

You can copy the section above, amended if necessary.

9.2 Data Preparation Code

Data Preparation Code ran without issues after the fix described in the Section [Replication](#).

9.3 Tables and Figures

Only a fraction of tables and figures was replicated due to replication interruption.

Figure/Table #	Program	Line Number	Reproduced?
Figure 1 (a), (b)	_data/figure_1_ab_table_a7a9.R	74; 112	Yes
Figure 1 (c), (d)	_data/figure_1_cd.R	70; 95	Yes
Figure 2 (a)	_data/figure_2_a.R	75; 109	Yes
Figure 2 (b)	_data/figure_2_b.R	94; 128	Yes
Table I	_model/table_i_and_ii.m	97	No
Figure 3	_model/figure_3_and_4.m	172; 248	No
Figure 4	_model/figure_3_and_4.m	318; 388	No
Figure 5	_model/figure_5.m	121; 184	No
Figure 6	_data/figure_6_data.R; _model/figure_6.m	62; 131	No
Table II	_model/table_i_and_ii.m	29	No
Figure 7	_model/figure_7.m	250; 265; 283; 301	No
Table III	_data/make_table3_data.do; _model/table_iii.m	197; 13-14; 72	No
Figure 8	_model/figure_8.m	244	No
Table IV	_model/table_iv.m	29	No
Table V	No new output; summary of Online Appendix C		No
Table VI	_model/table_vi.m	44	No
Table A.7	_data/figure_1_ab_table_a7a9.R	94	Yes
Table A.8	_data/table_a8.R	180	Yes
Table A.9	_data/figure_1_ab_table_a7a9.R	129	Yes
Table A.10	_data/table_a10.R	90	Yes
Table B.1	_model/_aux/_baseline/modelsMasterExcel.xlsx; _model/_aux/_kvar/modelsMasterExcel.xlsx		No
Table B.2	_data/table_b2_a.m; _model/table_bii.m	56; 104	No
Figure B.1	_model/figure_b1.m	53	No
Figure B.2	_model/figure_b2.m	147	No

Figure/Table #	Program	Line Number	Reproduced?
Table C.1	_model/table_ci.m	77	No
Table C.2	_model/table_cii.m	29	No
Figure C.1	_model/figure_c1.m	450	No
Table C.3	_model/table_ciii.m	137	No
Table C.4	_model/table_civ.m	30	No
Table C.5	_model/table_cv.m	28	No
Table C.6	_model/table_cvi.m	35	No
Table C.7	_model/table_cvii.m	35	No
Figure C.2	_model/figure_c2.m	229	No
Table C.8	_model/table_cviii.m	25	No
Figure C.3	_model/figure_c3.m	223	No
Table C.9	_model/table_cix.m	25	No
Figure C.4	_model/figure_c4.m	237	No
Table C.10	_model/table_cx.m	28	No
Table C.11	_model/_aux/_baseline/InfoDecomposition.m; _model/table_cxi.m	59; 80	No
Table C.12	_model/table_cxii.m	27	No
Figure D.1	_model/figure_d1.m	255	No
Table D.1	_model/table_di.m	44	No

9.4 In-Text Numbers

☐ In-text numbers not verified.

Replication did not identify code to reproduce paper in-text numbers.

In-text numbers	Program	Line Number	Reproduced?
p. 37: 2022 SCF wealth Gini is around 0.80	_model/table_iv.m	29	No

10 Classification

- ☐ full reproduction
☐ full reproduction with minor issues
☒ partial reproduction (see above)

- ☐ not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- ☐ None.
- ☐ **Discrepancy in output** (either figures or numbers in tables or text differ)
- ☒ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check
 - ☐ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☐ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☐ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☐ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements
- ☒ **Insufficient resources available to replicator**